

Part D: Recommendations Report

The SS1 project defined a clear business problem for STADIOEquities: the organisation has a large registered customer base but needs to understand which clients are at risk of becoming inactive. SS1 reported that 41% of registered accounts had never been funded and that dormancy within six months had increased from 22% to 31%. The proposed project therefore focuses on using information available before a prediction point to identify clients who are likely to become dormant within six months. The requested data includes onboarding, funding, digital behaviour, trading, products, client profiles, support and marketing engagement.

For SS2, a public customer-churn dataset was selected because STADIOEquities was not yet ready to release sensitive client information. The IBM Telco Customer Churn dataset contains 7,043 customer records and 21 raw columns and provides a binary churn outcome. IBM describes it as sample data for a fictional telecommunications company, with churn indicating whether a customer departed within the last month. The dataset is not identical to investment dormancy, so the SS2 results should be interpreted as a viability test of the modelling approach rather than as evidence about STADIOEquities clients.

Two models were specified. Logistic Regression was selected as an interpretable baseline because it estimates the probability of a binary outcome and provides a comparatively transparent reference model. Random Forest was selected because an ensemble of decision trees can capture nonlinear relationships and interactions. Both models use the same general preprocessing workflow: remove the customer identifier, convert TotalCharges to numeric, handle invalid records, encode categorical variables, create engagement-oriented features, and use class weighting. Evaluation uses accuracy, precision, recall, F1-score and ROC-AUC rather than accuracy alone.

The published benchmark used for the current draft reports that Logistic Regression achieved 83.26% accuracy, 82.59% precision, 84.29% recall, 83.43% F1-score and 91.96% ROC-AUC on the IBM dataset. The same study reports 84.73% accuracy, 85.20% precision, 84.07% recall, 84.62% F1-score and 93.86% ROC-AUC for Random Forest. These figures indicate a trade-off: Logistic Regression had slightly higher recall and much shorter training time, while Random Forest had stronger

accuracy, precision, F1-score and ROC-AUC in that published experiment (Mettle, Ayerko and Appiahene, 2026). The supplied code should be run before final submission so that these published benchmarks are replaced with the student's own empirical results.

Based on the cited benchmark, Random Forest is the model selected for the current viability recommendation because it provides stronger overall discrimination across accuracy, F1-score and ROC-AUC while maintaining similar recall to Logistic Regression. This conclusion is specific to the cited public-dataset experiment and should not be treated as a permanent conclusion for STADIOEquities. The real client data may have different class balance, feature distributions, temporal structure and operational costs.

The selected model can be improved in several ways. First, hyperparameter tuning should be performed using stratified cross-validation rather than relying on a single configuration. Second, probability calibration should be assessed because the business will use risk probabilities to prioritise clients. Third, class weighting and threshold selection should be evaluated against the cost of false negatives and false positives. Fourth, explainability should be added through permutation importance and, if approved, SHAP-style explanations. Finally, the production model should be monitored for data drift, changes in dormancy rates and deterioration in recall or calibration.

Adapting the model to the SS1 requested data requires a major change in the time structure of the features. The public dataset is a cross-sectional snapshot, whereas SS1 requires information available before a defined six-month dormancy outcome. STADIOEquities should therefore create a prediction date for each client and construct features only from earlier observations. Examples include days since last login, deposit recency, deposit frequency, withdrawal frequency, balance trend, number and value of trades, product adoption, support contacts, campaign engagement and onboarding completion. The target should be a consistently defined indicator showing whether dormancy occurred within six months after the prediction date.

The SS2 results are broadly consistent with the related literature in Part A. Mettle et al. (2026) show that Logistic Regression and Random Forest are useful classical

approaches for the IBM dataset and that Random Forest can provide stronger overall discrimination. Sikri et al. (2024) emphasise class imbalance and show the usefulness of balancing strategies and ensemble methods. Ahmad, Jafar and Aljoumaa (2019) demonstrate that feature engineering, feature selection and AUC-based evaluation are important in churn modelling. These findings support the SS1 emphasis on integrating behavioural data and constructing features before model training. However, the studies differ in datasets, time periods and modelling pipelines, so their numerical results should not be transferred directly to STADIOEquities.

SS2 demonstrates that a customer-attribution prediction workflow is technically viable using public data, while also identifying the limitations that must be addressed before client deployment. The next stage should be to obtain the STADIOEquities datasets specified in SS1, define the six-month dormancy target precisely, construct temporally valid features, tune and calibrate the candidate models, and evaluate them against business costs. The public-dataset experiment should therefore be treated as a controlled proof of concept rather than as a finished STADIOEquities production model.

References

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- Sikri, A., Jameel, R., Idrees, S.M. and Kaur, H. (2024). Enhancing customer retention in telecom industry with machine learning driven churn prediction. *Scientific Reports*, 14, 13097. <https://doi.org/10.1038/s41598-024-63750-0>
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